

Tudor Andrei Dumitrascu *AI Architect*

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SKILLS

AI/ML: LangChain, Pydantic AI, Claude SDK, DSPy, PyTorch, Scikit-Learn, NVIDIA Triton, NLP, CV, DL; **Backend:** Python, FastAPI, NestJS, Litestar, RESTful APIs; **Data:** Pandas, MongoDB, PostgreSQL, Redis, RabbitMQ, SurrealDB; **Frontend:** Typescript, Next.js, React, Remix, TailwindCSS; **Cloud/DevOps:** Docker, AWS, Azure, CI/CD, Prometheus, Grafana, Nix; **Languages:** Romanian, English, German

WORK EXPERIENCE

Member of Technical Staff Footprints AI	Jun 2026 — Present <i>Bucharest, Romania</i>
- Migrated the core codebase from MongoDB to PostgreSQL, moving the retail-media platform onto a relational data model - Built an integration plugin platform that extends the web SDK, so new integrations ship as plugins instead of changes to the core - Improving the data platform that runs ETL across tenant deployments	
CTO Pythia Sociodynamical Technologies	Mar 2026 — Present <i>Bucharest, Romania</i>
- Setting technical direction and running engineering operations - Building tooling for AI engineering: eval harnesses, agent infrastructure, developer workflow - Delivering client AI projects end to end, plus AI consultancy and workshops	
CTO Flow Copilot	Feb 2026 — Present <i>Bucharest, Romania</i>
- Built Flow Copilot, an AI task-management app for people with ADHD (iOS/Android/web) - Owned the voice-to-task pipeline end to end: speech transcription, LLM-based task extraction and editing, and long-form audio processing - Built the Bun/Elysia backend, Convex realtime sync, and Google Calendar integration - Built the eval harness used to measure AI output quality - Set up the team's CI/CD, secrets management, and monorepo tooling	
Auditor Cert Sud	Jan 2026 — Present <i>Bucharest, Romania</i>
- Performing ISO/IEC 42001 (AI management systems) and ISO/IEC 27001 (information security) audits for client organizations - Assessing AI governance, risk controls, and documentation against certification requirements	
AI Engineer CogniSync	Sep 2025 — Mar 2026 <i>Bucharest, Romania</i>
- Developed core platform feature that ingested complex legal playbooks and templates to create structured rule systems - Engineered AI system capable of processing and understanding large legal documents, reducing manual review time - Created document parsing engine that identifies and extracts placeholder items from templates - Prototyped document-aware Q&A system using Retrieval-Augmented Generation (RAG) for legal queries - Implemented and compared two RAG approaches: Google Vertex AI integration and custom system using Qdrant - Applied modern LLM techniques including RAG and prompt engineering for legal domain applications	
Co-Founder & CTO Pythia Sociodynamical Technologies	Jun 2022 — Sep 2025 <i>Bucharest, Romania</i>
- Architected NVIDIA Triton-based ML inference pipeline that processes 1,000+ news articles per second with <100ms latency, reducing infrastructure costs by 35% - Engineered LLM & DSPy powered data extraction with pipeline with intelligent content hashing, increasing data extraction accuracy by >40% while reducing inference costs - Built distributed news scraper with Scrapy and RabbitMQ that integrated 41+ sources in one week, processing 500K+ articles daily with 99.5% uptime	

- Developed multi-tenant SaaS platform leveraging Next.js and SurrealDB for automated KYC/AML compliance workflows, serving multiple banks and crediting companies
- Designed RESTful API architecture with FastAPI, enabling seamless microservices integration and cross-platform data exchange
- Designed custom algorithms generating plausible trading scenarios from real-world Time-Series financial data
- Developed bias detection algorithms on Time-Series events for financial risk assessment
- Delivered efficient Named Entity Recognition and Sentiment Analysis solutions for financial text processing
- Implemented Graph Database solution using SurrealDB, optimizing complex relationship queries and improving data retrieval performance by 60%
- Deployed comprehensive observability stack with Prometheus and Grafana, maintaining 99.9% system uptime across production environments

IT Administrator

Cert Sud SRL

Sep 2020 — Sep 2022

Bucharest, Romania

- Managing IT infrastructure
- Website Designer

Research & Development Intern

Continental Powertrain Engineering

Jun 2019 — Aug 2019

Timisoara, Romania

- CAN Bus programming with Arduino Uno

EDUCATION

University of Bucharest

M.Sc. Artificial Intelligence

Bucharest, Romania

2020 — 2022

Julius Maximilian University of Würzburg

Erasmus+, Artificial Intelligence

Würzburg, Germany

2021 — 2021

Polytechnics University of Bucharest

B.Eng. Applied Electronics

Bucharest, Romania

2016 — 2020

Åbo Akademi University

Erasmus+, Computer Science

Turku, Finland

2019 — 2020

Academy+Plus

Coding Bootcamp

Bucharest, Romania

Sep 2018 — Jan 2019

Colegiul Național Bilingv "George Coșbuc"

Bacalaurate Diploma

Bucharest, Romania

Sep 2012 — Jul 2016

RESEARCH

Patronizing and Condescending Language Detection, SemEval2022 Shared Task (ACL) ([Link](#)) Nov 2022 — Feb 2023

- Developed advanced BERT-based models for binary classification and multi-label multi-class text classification
- Implemented machine learning algorithms using bag-of-words models for patronizing language detection
- Published research addressing linguistic bias detection in natural language processing

Detecting Early Signs of Pathological Gambling using ML and DL models with dataset chunking, CLEF eRisk 2022 ([Link](#)) Nov 2021 — May 2022

- Applied machine learning and deep learning models with dataset chunking techniques
- Developed early warning systems for pathological gambling behavior detection

PROJECTS

Opticale.AI, Detection and Tracking system using AI

Sep 2019 — Mar 2020

- Developed real-time boat detection and tracking system using computer vision and AI
- Successfully commercialized project through acquisition by maritime autonomous solutions company

CERTIFICATES

ISO 9001:2015 Auditor/Lead Auditor (CQI/IRCA Certified Course)

RINA (Jul 2026)

ISO/IEC 42001:2023 Internal Auditor

TÜV Thüringen Karpát (Apr 2026)

ISO/IEC 27001:2022 External Auditor

CERTINSPECT Register (Feb 2023)

Deep Learning Specialization

DeepLearning.AI (2021)

Sequence Models

DeepLearning.AI (2021)

Convolutional Neural Networks

DeepLearning.AI (2020)

Improving Deep Neural Networks

DeepLearning.AI (2020)

Structuring Machine Learning Projects

DeepLearning.AI (2020)

Neural Networks and Deep Learning

DeepLearning.AI (2020)

Introduction to TensorFlow for AI

DeepLearning.AI (2019)

ORGANIZATIONS

Liga Studentilor Electronisti, Volunteer

2016 — 2019 | Bucharest, Romania

AWARDS

Bronze Medal, MobilPRO

17/03/2019

Fifth Place, 2NHack

02/11/2019

INTERESTS

Science-Fiction, Language Learning, Sports, Cardistry, Philosophy